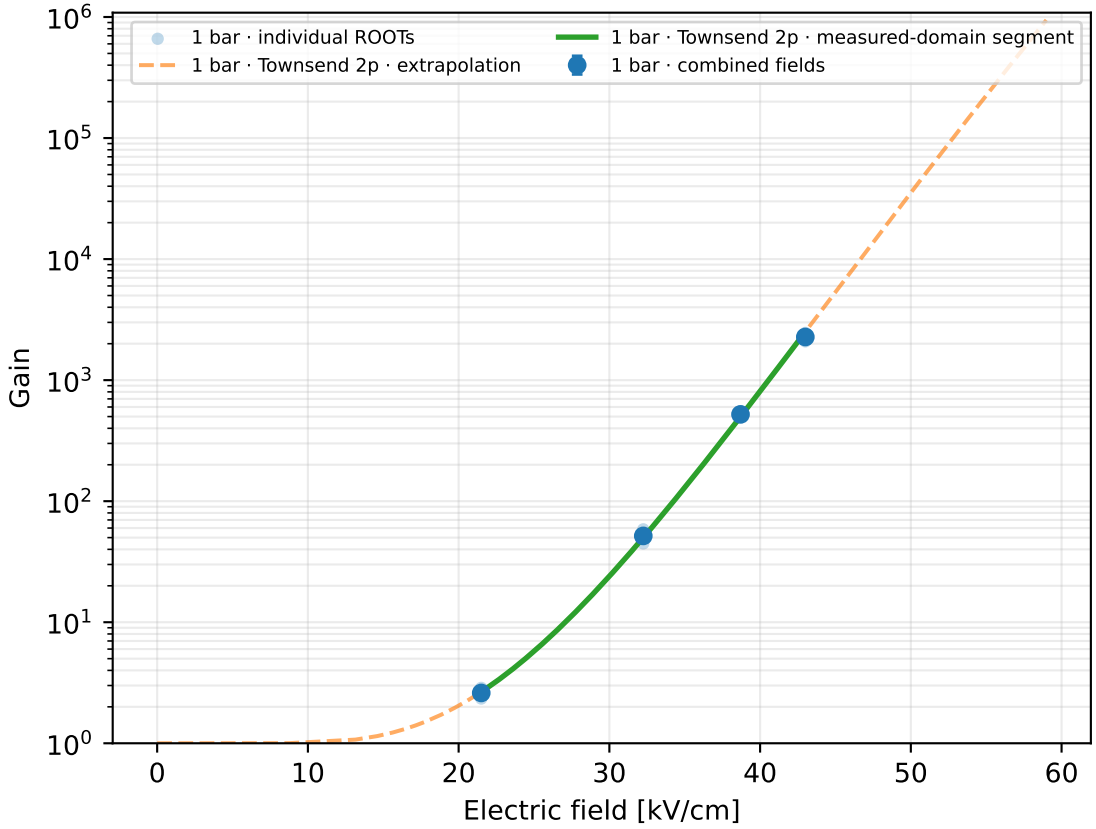
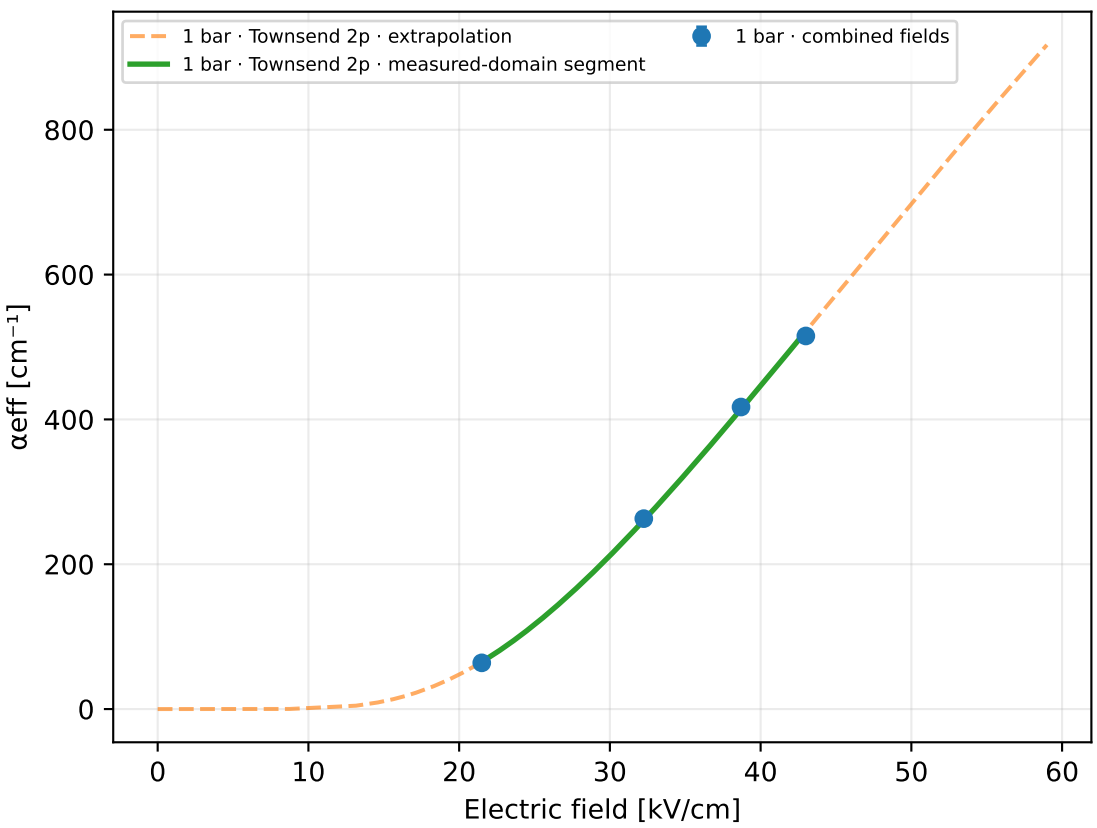
ArCF4 · Ar 85 % / CF_4 15 % · gap 0.15 mm

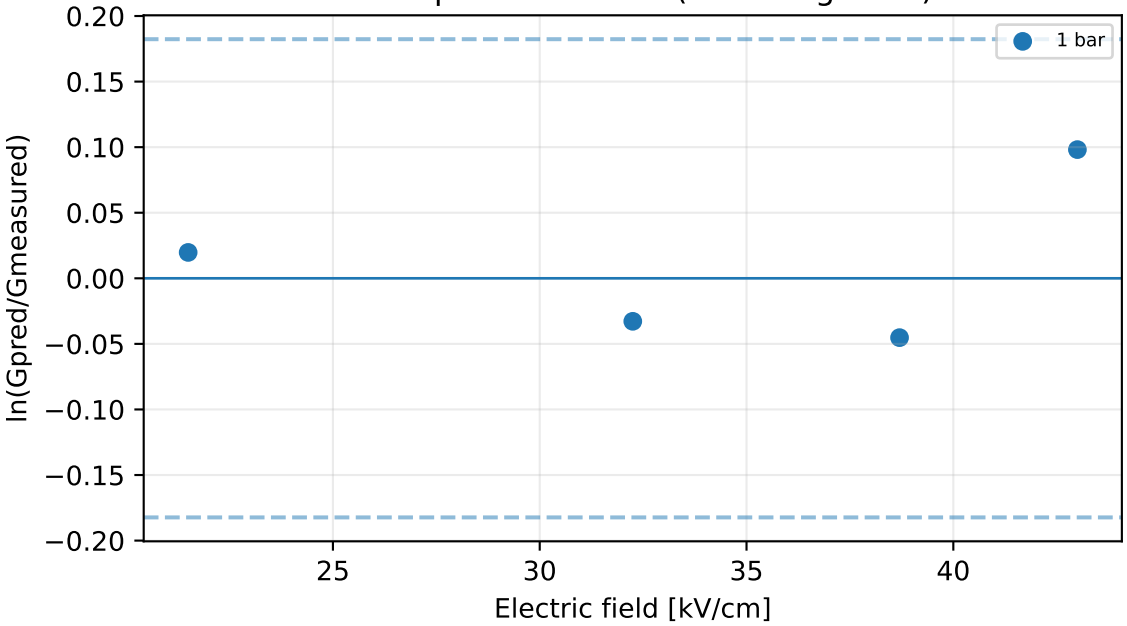
Gain vs electric field



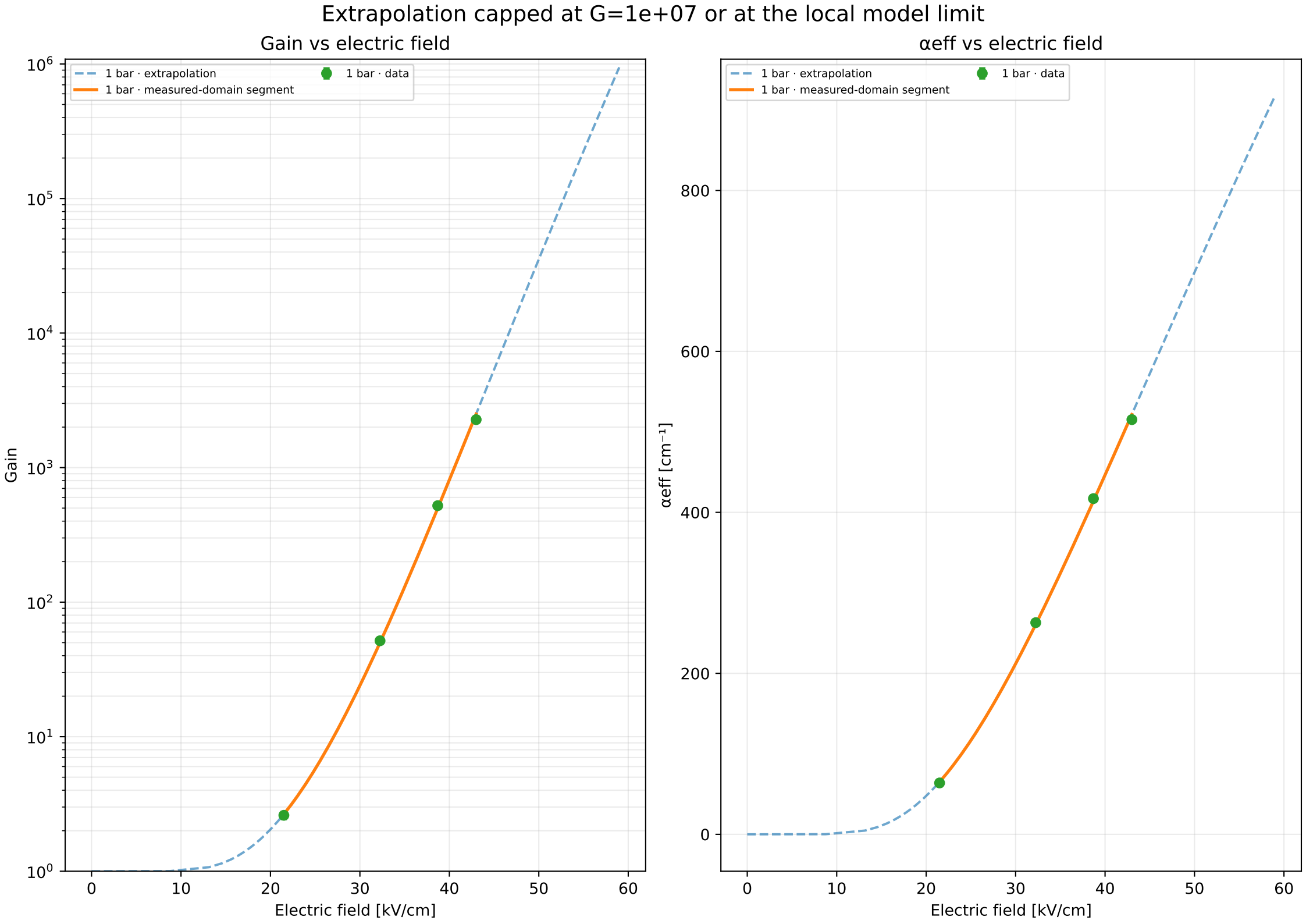
Effective Townsend coefficient vs electric field



Gain-space residuals (±ln 1.2 guides)




```
ROOT runs: 8
Unique physical points: 4
Pressures: 1 bar
Selected model: Townsend 2p
Status: rejected_poor_cross_validation
Training RMSE: ×1.013
Cross-validation median: unavailable
Cross-validation worst: unavailable
Reduced-field range: 21.5–43 kV cm⁻¹ bar⁻¹
A=4179.2, B=89.467, m=0, n=1
Covariance condition: 229
Warnings: poor_cross_validation, large_cross_validation_outlier
1 bar display limit: G=9.44e+05 (local inversion limit)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
1 bar · extrapolation
1 bar · measured-domain segment
1 bar · data
Gain
Electric field [kV/cm]
αeff vs electric field
αeff [cm⁻¹]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
1	21.5	21.5	2.605	0.178	63.8288	200	2
1	32.25	32.25	51.655	3.2	262.972	200	2
1	38.7	38.7	521.375	29.3	417.098	200	2
1	43	43	2272.325	126	515.237	200	2

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-18.6	—	—	no	poor cross validation, large cross validation outlier